

Deep Learning for Image Analysis

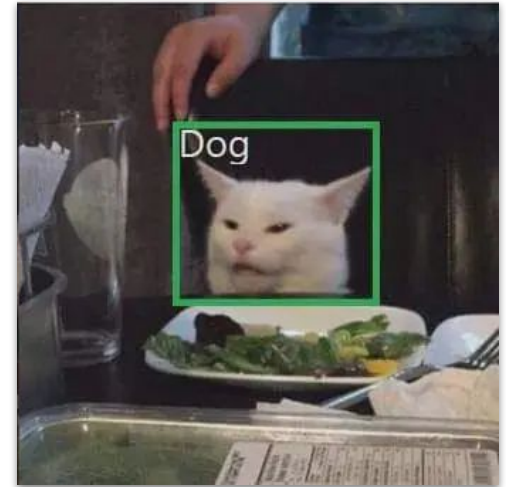
Mehyar MLAWEH

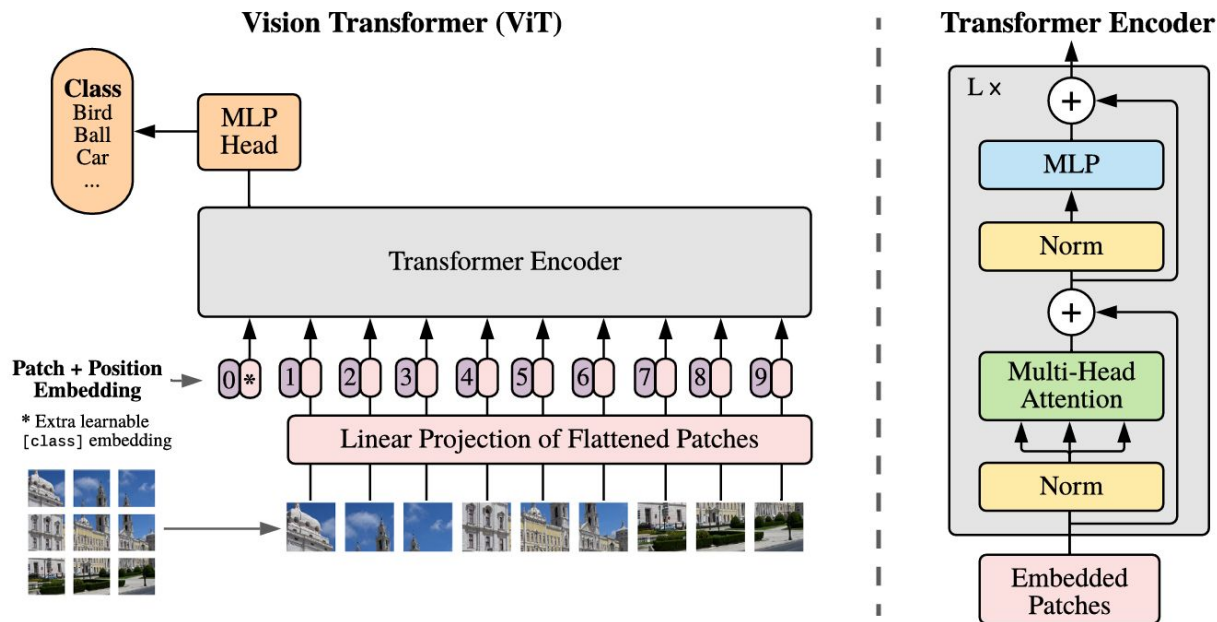


Master BDIA - Dauphine Tunis
PhD student - Université PSL

1. Computer Vision Foundations
2. CNNs
3. Transfer Learning
4. Vision Transformers
5. Object Detection
6. Segment Anything Model (SAM)
7. Final Project

Lecture 5 : Object Detection





The goal of object detection is to localize objects in an image and tell their **class**

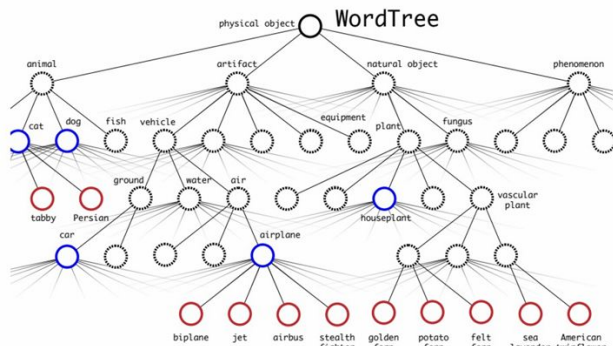
Localization: place a tight bounding box around object



The goal of object detection is to localize objects in an image and tell their **class**

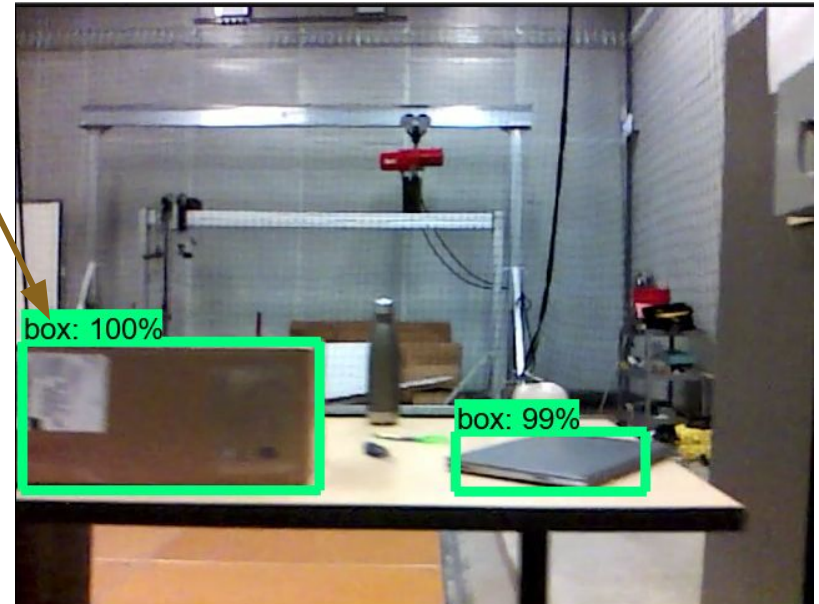
Localization: place a tight bounding box around object

Can scale up to **many classes** using **hierarchical tree** of visual concepts

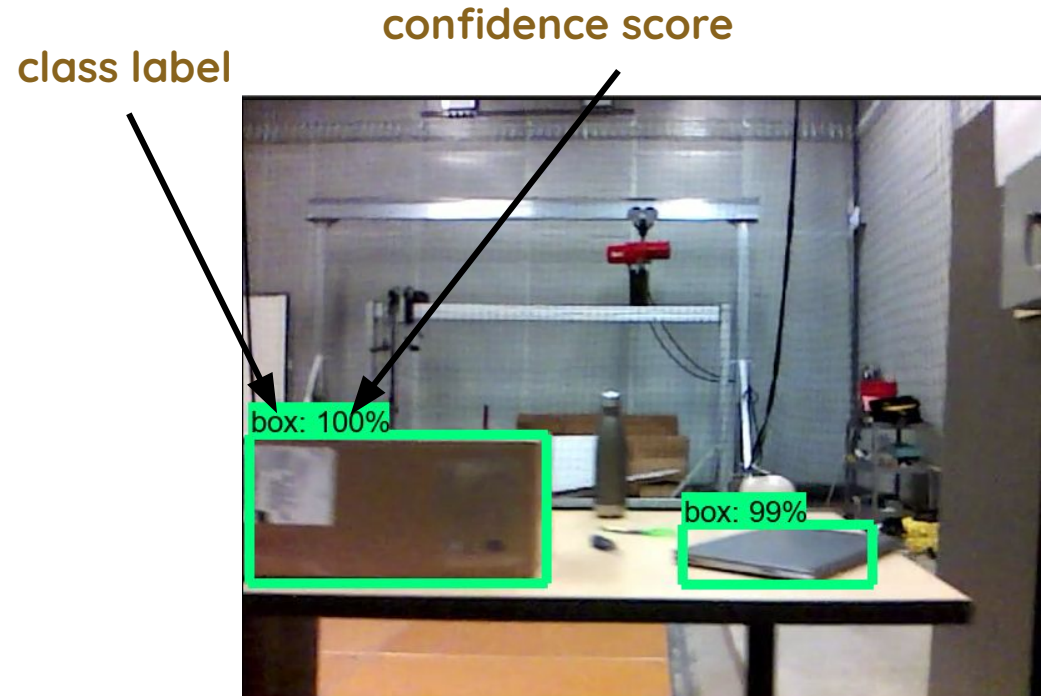


For each detected object, the model outputs:

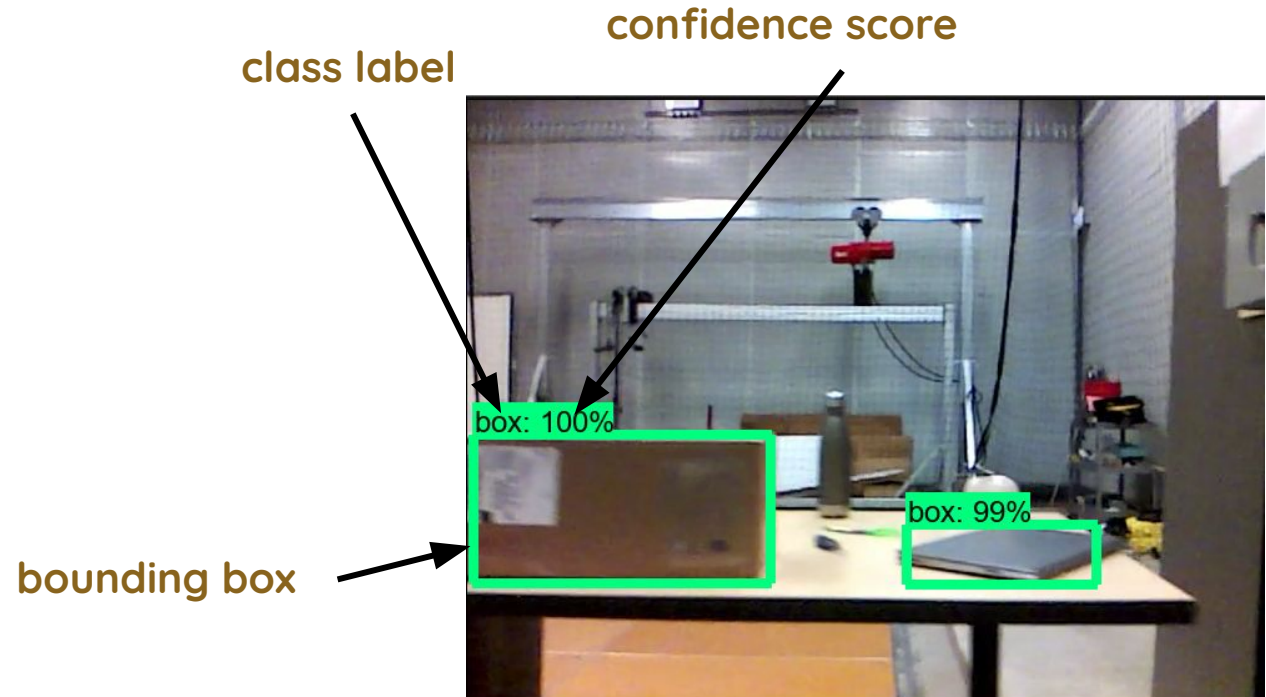
class label



For each detected object, the model outputs:

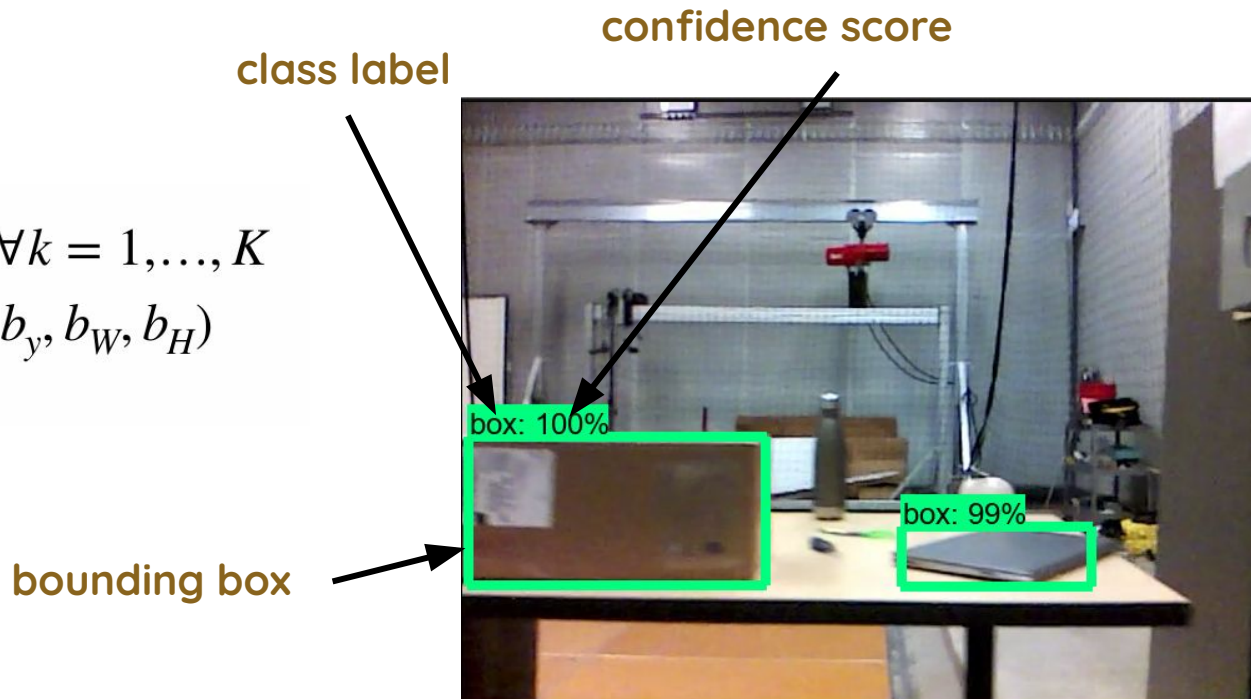


For each detected object, the model outputs:



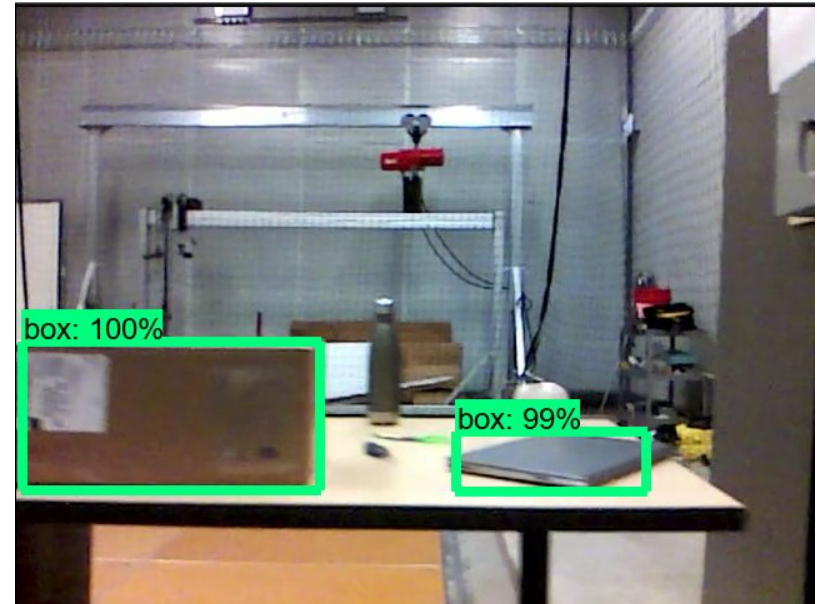
For each detected object, the model outputs:

$$\forall \text{ object} : \begin{cases} P(C_k) & \forall k = 1, \dots, K \\ \mathbf{b} = (b_x, b_y, b_W, b_H) \end{cases}$$



Measure performance

How good is a predicted bounding box ?



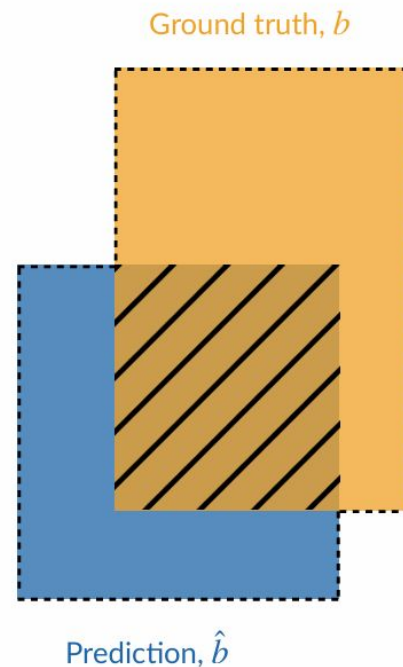
Measure performance

How good is a predicted bounding box ?

We calculate Intersection over Union:

$$\text{IoU}(\hat{b}, b) = \frac{\text{area}(\hat{b} \cap b)}{\text{area}(\hat{b} \cup b)}$$

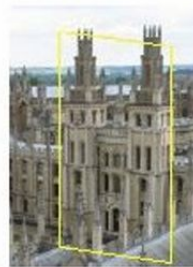
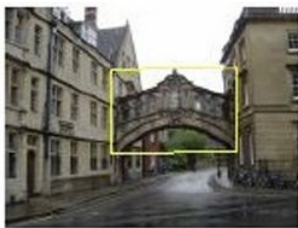
Prediction correct if $\text{IoU} > 0.5$



[Slide credit: Vito Dichio]

Major Difficulties :

- Objects appear at **different** scales, viewpoints, lightings ..



[Source: J. Sivic, slide credit: R. Urtasun]

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Major Difficulties :

- Objects appear at **different** scales, viewpoints, lightings ..
- Multiple objects can overlap



[Source: J. Sivic, slide credit: R. Urtasun]

Major Difficulties :

- Objects appear at **different** scales, viewpoints, lightings ..
- Multiple objects can overlap
- Expensive annotation



[Source: J. Sivic, slide credit: R. Urtasun]

Slide window and ask a classifier: “Is sheep in window or not?”



Confidence : 0.1

[Source: R. Urtasun]

Slide window and ask a classifier: “Is sheep in window or not?”



Confidence : -0.2

[Source: R. Urtasun]

Slide window and ask a classifier: “Is sheep in window or not?”



Confidence : -0.1

[Source: R. Urtasun]

Slide window and ask a classifier: “Is sheep in window or not?”



Confidence : 1.5

[Source: R. Urtasun]

Slide window and ask a classifier: “Is sheep in window or not?”



Confidence : 0.5

[Source: R. Urtasun]

Slide window and ask a classifier: “Is sheep in window or not?”



Results

Confidence

-0.1

0.2

-0.1

...

1.5

...

0.5

0.2

-0.1

[Source: R. Urtasun]

Downsides

- Computationally expensive
- Inaccurate bounding box : None of the squares could match the actual size of the object perfectly; each square may be smaller or larger.
- More than one object within same grid

[Source: R. Urtasun]

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[Source: R. Urtasun]

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Group pixels into object-like regions



[Source: R. Urtasun]

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Generate many different regions



[Source: R. Urtasun]

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Generate many different regions



[Source: R. Urtasun]

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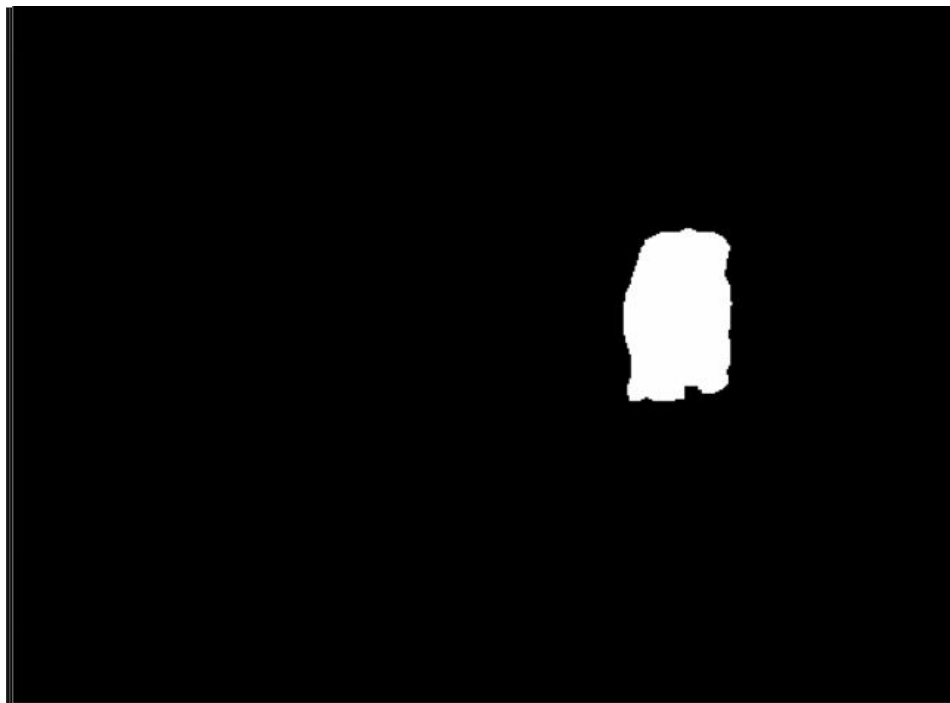
Generate many different regions



[Source: R. Urtasun]

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Generate many different regions



[Source: R. Urtasun]

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Generate many different regions



[Source: R. Urtasun]

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The hope is that at least a few will cover real objects



[Source: R. Urtasun]

Select a region



[Source: R. Urtasun]

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Crop out an image patch around it, throw to classifier (e.g., Neural Net)



classifier

'Dog' or not ?

Confidence : -0.25

[Source: R. Urtasun]

Crop out an image patch around it, throw to classifier (e.g., Neural Net)



classifier

'Dog' or not ?

Confidence : -0.25

[Source: R. Urtasun]

Crop out an image patch around it, throw to classifier (e.g., Neural Net)



classifier

‘Dog’ or not ?

Confidence : 1.5

[Source: R. Urtasun]

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Crop out an image patch around it, throw to classifier (e.g., Neural Net)



classifier

'Dog' or not ?

Confidence : 1.5

A DOG !!

[Source: R. Urtasun]

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We generate many candidate regions that may contain objects, then
classify each region.

Downsides

- Too many candidate regions
- Multi-stage pipeline: image → generate proposals → crop/resize regions → classify each region → refine boxes → remove duplicates
- Slow for real-time applications

[Source: R. Urtasun]

It uses region proposals, then applies a CNN to each proposed region to extract features and classify the object



CNN

‘Dog’ or not ?

A DOG !!

Confidence : 1.5

[Source: R. Urtasun]

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Still slow !!



CNN

‘Dog’ or not ?

A DOG !!

Confidence : 1.5

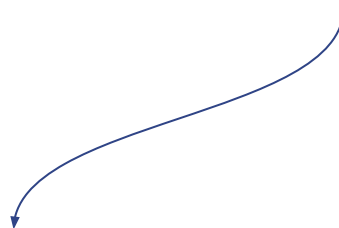
[Source: R. Urtasun]

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Still slow !!



because the CNN is applied many times



CNN

‘Dog’ or not ?

A DOG !!

Confidence : 1.5

[Source: R. Urtasun]

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So the main problem is the number of iteration we are doing



CNN

'Dog' or not ?

A DOG !!

Confidence : 1.5

[Source: R. Urtasun]

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So the main problem is the number of iteration we are doing

We only need to look once to make it faster



CNN

'Dog' or not ?

A DOG !!

Confidence : 1.5

[Source: R. Urtasun]

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Segment Anything Model

Books

- Computer Vision: Models, Learning, and Inference (2012). <https://amzn.to/2rxrdOF>
- Programming Computer Vision with Python (2012). <https://amzn.to/2QKTaAL>
- Multiple View Geometry in Computer Vision (2004). <https://amzn.to/2LfHLE8>
- Computer Vision: Algorithms and Applications (2010). <https://amzn.to/2Lclt4J>
- Computer Vision: A Modern Approach (2002). <https://amzn.to/2rv7AqJ>
- Deep Learning for Computer Vision : Image Classification, Object Detection and Face Recognition in Python(2019). <https://machinelearningmastery.com/deep-learning-for-computer-vision/>
- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow — Aurélien Géron (2019)
<https://amzn.to/3uZbN6Z>
- Hands-On Large Language Models - Jay Alammar & Grootendorst

Articles & Online Resources

- Krizhevsky, Sutskever, Hinton (2012) — ImageNet Classification with Deep CNNs (AlexNet — deep learning breakthrough)
- Improved Texture Networks: Maximizing Quality and Diversity in Feed-forward Stylization and Texture Synthesis, 2017
- “Computer vision,” Wikipedia. https://en.wikipedia.org/wiki/Computer_vision
- “Machine vision,” Wikipedia. https://en.wikipedia.org/wiki/Machine_vision
- “Digital image processing,” Wikipedia. https://en.wikipedia.org/wiki/Digital_image_processing

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- <https://medium.com/data-reply-it-datatech>
- <https://huggingface.co/blog/RDTvlokup/quand-l-ia-apprend-de-l-experience-comme-toi>
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- <https://jalammar.github.io/illustrated-transformer/>
- <https://sh-tsang.medium.com/review-vision-transformer-vit-406568603de0>
- Laurier, Linda. (2025). Vision Transformers: From Patch-Based Processing to Visual Intelligence.
- “Training Convolutional Neural Networks: What Is Machine Learning” Analog Devices.
<https://www.analog.com/en/resources/analog-dialogue/articles/training-convolutional-neural-networks-what-is-machine-learning-part-2.html>
- Roth, Peter M.. (2011). On-line Conservative Learning.